Heard v0.1: A Korean Long-Term Memory Benchmark and On-Device Retrieval Pipeline for Solo-Business Monologue

Chanyoung Kim · NLP Term Project Part 2 · 2026-04-28

Website	https://cykim05.github.io/
Code	https://github.com/cykim05/heard
Dataset	https://huggingface.co/datasets/chanyoungkim/heard-bench

Summary

Heard (Part 1 NLP proposal) is an on-device Korean LLM assistant that reflects a user's past self back at them at decision moments. Its three architectural pillars are MIC (speech-to-text, implemented as a sounddevice + faster-whisper pipeline but not evaluated in v0.1), NODE (domain-specific typed memory), and MIRROR (reflective response). Part 2 asks whether a lightweight on-device memory pipeline yields measurable gains for a small Korean LM on nightly-self-talk questions, and whether a reflective prompt policy beats a generic advisory one.

Evaluating this requires a long-term memory benchmark in Korean, covering monologue (not dialogue), in the solo-business domain. No existing benchmark — LongMemEval, PerLTQA, or LoCoMo — satisfies all three criteria, so we constructed heard-bench: 270 items across three tracks (en_subset, ko_translated, ko_native), generated from a 2,046-utterance synthetic corpus over three solo-business personas, adversarially filtered against no-memory baselines, 4-gate auto-validated, and author-reviewed. The dataset is released on HuggingFace under CC-BY-4.0.

On the resulting benchmark, dense retrieval lifts Kanana-2.1B's ko_native pass rate from 4.7% (no memory) to 10.9% (retrieval) to 15.6% (oracle), a 3.3× end-to-end gain. Pass rate decays monotonically across language tracks (ko_native 10.9%, ko_translated 5.0%, en_subset 0.0%), establishing Korean-native data as necessary rather than optional. A reflective policy dominates an advisory baseline on emotional attunement (82 of 96 wins) and open-question framing (88 of 96) across 96 pairwise judge decisions.

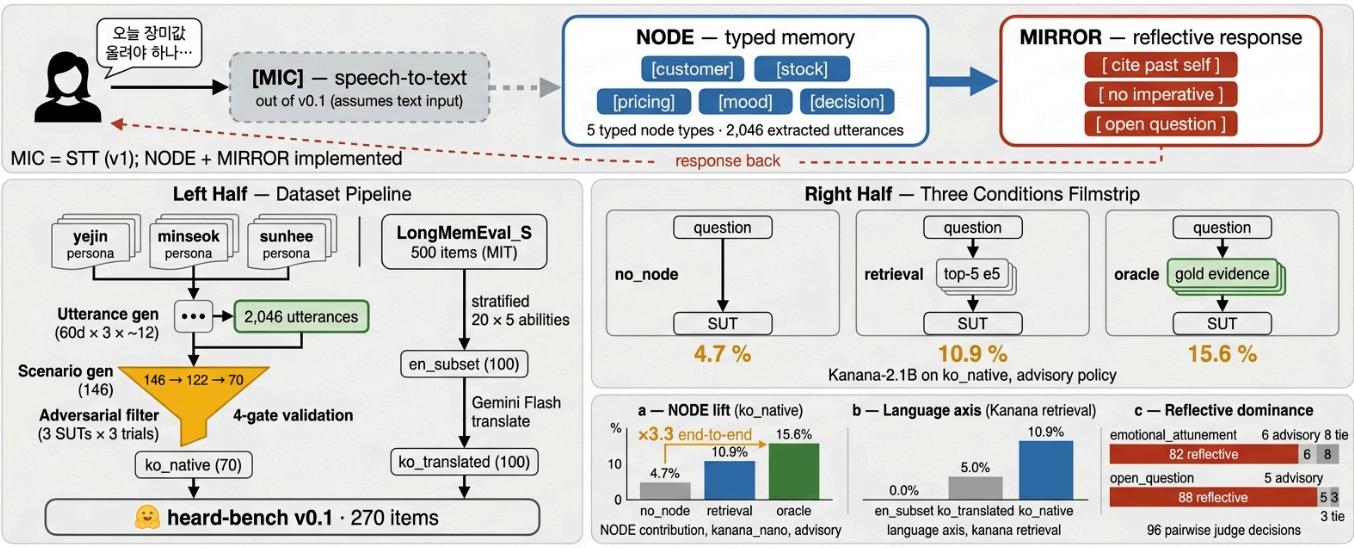


Figure 1: Heard v0.1 Pipeline & Headline Result Overview

{#fig:overview}

1 Introduction

1.1 Background

Heard is the Part 1 NLP proposal for an on-device Korean LLM assistant. Its architecture has three pillars. MIC handles always-off tap-to-talk speech-to-text; it is implemented in v0.1 but not evaluated in Part 2 (§2.1). NODE is the domain-specific typed memory built over utterances. MIRROR produces reflective responses that quote the user's past words and end with an open question rather than giving advice.

The proposal targets Korea's 1.4M-member solo-business community, where nightly self-talk arrives from people who have no coworkers, no staff, and no one to check in with. Part 2 asks two empirical questions: (1) does a lightweight retrieval-based memory pipeline yield measurable gains for a small Korean on-device LM on nightly self-talk questions, and (2) does a reflective prompt policy beat a generic advisory policy when memory is available?

1.2 The benchmark gap

Answering these empirical questions requires a long-term memory benchmark meeting three criteria: (a) Korean, (b) monologue (nightly self-talk, not dialogue), and (c) solo-business decision-making. A survey of recent long-term memory benchmarks returns no candidate that satisfies all three.

Benchmark	**Language**	**Format**	**Domain**
LongMemEval (Wu et al., ICLR 2025)	EN	user–AI dialogue	generic chat
PerLTQA (Du et al., 2024)	ZH / EN	human–human dialogue	social events
LoCoMo (Maharana et al., 2024)	EN	long dialogue	generic personal

We therefore construct **heard-bench**: a 270-item benchmark across three tracks. The construction pipeline is: (1) generate a 2,046-utterance synthetic corpus over three solo-business personas (florist, roaster-café owner, one-chair hair salon), (2) compose scenarios with today-questions and evidence utterances, (3) adversarially filter against no-memory baselines (3 SUTs × 3 trials, keeping only items where all fail), (4) 4-gate auto-validate (evidence consistency, history-only fail, cross-generator duplication, ambiguity score), and (5) author-review the survivors for naturalness and gold-answer correctness. The dataset is released on HuggingFace under CC-BY-4.0.

1.3 Contributions

1. **Dataset construction as necessary infrastructure** (§2.2). We construct, adversarially filter, 4-gate validate, and author-review heard-bench, a 270-item set across en_subset, ko_translated, and ko_native tracks, and release it on HuggingFace under CC-BY-4.0. No existing long-term memory benchmark meets our three criteria of Korean / monologue / solo-business.

2. **Empirical NODE lift** (§3.1). Dense retrieval lifts Kanana-2.1B's ko_native pass rate from 4.7% (no memory) to 10.9% (retrieval) to 15.6% (oracle), a 3.3× end-to-end gain over the memoryless baseline.

3. **Language axis decay confirms Korean-native necessity** (§3.2). Kanana's retrieval pass rate decays monotonically: ko_native 10.9% → ko_translated 5.0% → en_subset 0.0%. Korean-native data is necessary, not optional, for a Korean on-device assistant.

4. **Reflective policy dominates on emotional and open-question axes** (§3.3). Across 96 pairwise judge decisions per rubric, reflective beats advisory on emotional attunement 82/96 and on open-question framing 88/96, supporting the Part 1 MIRROR thesis.

2 Methods

All code, configurations, and experiment artifacts are available at **https://github.com/cykim05/heard** (MIT). The dataset is published at **https://huggingface.co/datasets/chanyoungkim/heard-bench** (CC-BY-4.0). Per-call API logs under `experiments/_api_log/` and per-run folders under `experiments/<run_id>/` enable full reproducibility; diskcache keyed on `(model, messages, seed)` makes a cold rerun free.

2.1 MIC — recording and transcription

Heard's first pillar turns spoken nightly monologue into text before NODE ingests it. We implement it as a lightweight tap-to-talk recorder plus a Whisper-family transcriber; the code ships with v0.1 but is not exercised by the evaluation in §3.

Recorder (`src/mic/recorder.py`). A `sounddevice` wrapper captures 16 kHz mono PCM in float32, which is the format Whisper-family models expect natively. Two usage modes are supported: `record_fixed(seconds)` for scripted capture, and a `Recorder` context manager for tap-to-talk (start, wait for the user, stop). The captured array is either passed directly to the transcriber or written to a 16-bit WAV via `save_wav`.

Transcriber (`src/mic/transcribe.py`). A thin `faster-whisper` wrapper (CTranslate2 backend), defaulting to the Whisper `small` model (244 M parameters). Language is fixed to Korean; VAD filtering and beam search are exposed as kwargs. Heavy model loading is lazy, so importing the module is cheap even in environments that never run STT. `transcribe()` returns a single concatenated transcript; `transcribe_segments()` returns per-segment timestamps for later utterance splitting.

Integration. `scripts/00_record_and_transcribe.py` is the reference integration: it records via `Recorder`, optionally saves a WAV, and routes the audio through `Transcriber`. The script ships as a working demo rather than a v0.1 evaluation target, since measuring WER on Korean casual monologue requires a dedicated audio benchmark that we leave to v2. All subsequent evaluation in §3 operates on gold utterance text so that the memory-pipeline question is isolated from STT error.

2.2 Dataset construction

The heard-bench construction pipeline has five stages: corpus generation, scenario composition, adversarial filtering, 4-gate auto-validation, and author review. We describe each in turn.

Personas. Three synthetic Korean solo-business owners anchor the corpus: Yejin (florist, Mapo), Minseok (roaster-café, Seongsu), and Sunhee (one-chair hair salon, Hongje). Each persona card specifies regulars, stock or services, recurring stressors, and timestamped historical anchors (`days_ago`-indexed past events) so that temporal-reasoning and multi-session items have grounded evidence. The persona set is synthetic to avoid privacy concerns; all names and establishments are fictitious.

Utterance corpus. One generator call per (persona, day) block across 60 days produces 12 utterances per day on average, rotating across `anthropic/claude-haiku-4.5`, `openai/gpt-4o-mini`, and `google/gemini-2.5-flash` so that no single generator's fingerprint dominates. 43% of the resulting utterances reference a persona historical anchor, and these references become retrieval gold hooks in later stages.

Persona	Role / location	Days	Utterances	Historical-ref share
yejin_florist	Florist, Mapo	60	686	43%
minseok_cafe	Roaster-café, Seongsu	60	687	43%
sunhee_hair	One-chair hair salon, Hongje	60	673	43%
Total			**2,046**	**881 / 2,046**

Three tracks. heard-bench is organized into three tracks.

Track	Items	Language	Haystack	Source	License
`en_subset`	100	EN	EN	LongMemEval_S stratified (20 per ability)	MIT
`ko_translated`	100	KO Q + answer	EN (preserved)	Gemini 2.5 Flash translation	CC-BY-4.0
`ko_native`	70	KO	KO (our corpus)	persona-driven gen + adversarial filter	CC-BY-4.0
Total	**270**				

Ability distribution (ko_native). The native-track items are stratified by the six long-term-memory abilities we target. IE, MR, KU, TR, and ABS follow LongMemEval's taxonomy; REFL (reflective quality) is specific to Heard.

Ability	Count	What it tests
IE	19	Specific fact recall

Ability	Count	What it tests
TR	15	Temporal reference resolution
KU	13	Knowledge update (latest state)
ABS	10	Abstention (never mentioned)
MR	7	Multi-session aggregation
REFL	6	Reflective response quality (LLM-judged)

Validation pipeline. ko_native candidates pass through four stages before inclusion:

1. **Adversarial filtering**: 150 candidates generated → 146 successfully parsed → 122 survived (all three no-memory baselines, Kanana-2.1B / Qwen-1.5B / Qwen-3B, fail across three trials each). This ensures that memory is structurally necessary to solve the retained items.
2. **4-gate auto-validation**: evidence-answer consistency, history-only fail reproduction, cross-generator 4-gram overlap < 15%, and ambiguity score ≥ 3/5 from a third-party LLM rater (ADR 0004).
3. **Author review**: the 70 surviving items were reviewed by the author for naturalness, evidence alignment, and gold-answer correctness. Minor edits to gold substrings and fluency fixes were applied as needed.
4. **Final count**: 70 ko_native items, shorter than the originally planned 100. The rejection rate at Gate 4 is discussed in §4.5.

Algorithm 1 summarises the adversarial filter. Any candidate that is solvable without memory by any of the three SUTs in any of three trials is discarded; only candidates where memory is structurally required survive.

```
Algorithm 1 — Adversarial filter
---------------------------------------------------------
input  : candidate set C, SUT pool M = {m1, m2, m3},
         trials T = 3
output : hard subset H ⊆ C

H ← ∅
for each candidate c ∈ C:
    any_pass ← False
    for each m ∈ M:
        for t in 1..T:
            r ← m.generate(c.question)      // no memory
            if gold_criterion(r, c.gold):
                any_pass ← True
                break
        if any_pass: break
    if not any_pass:
        H ← H ∪ {c}
return H
```

ADR 0003 summary. Track B keeps only question and gold_answer in Korean; haystack stays English, because full-session translation exceeded Gemini-Flash's output token cap. Chunked-session translation is a concrete v2 path.

ADR 0004 summary. Gate 1 was demoted to advisory after a scenario-generation regression produced false negatives that did not correlate with SUT evaluation value.

2.3 Memory pipeline

Retriever. We use `intfloat/multilingual-e5-small` (118 M parameters), mean-pool the final hidden states, L2-normalize, and take cosine top-5. The index is per-persona for ko_native (~680 utterances each) and per-item for en_subset and ko_translated (~50 sessions per item).

2.4 SUTs (systems under test)

Logical	HF id	Params	Quant
kanana_nano	`kakaocorp/kanana-1.5-2.1b-instruct-2505`	2.09 B	fp16
qwen25_3b	`Qwen/Qwen2.5-3B-Instruct`	3.09 B	fp16

Both SUTs run on a single NVIDIA L40S 48 GB, sequentially loaded. Models excluded from v0.1: EXAONE-3.5 and HyperCLOVA-X-SEED (requiring additional license acceptance), Gemma-2 (gated), the int4 quantization axis, and Kanana-8B as reference ceiling. All are concrete v2 items.

2.5 MIRROR policies

We implement two policies in `src/mirror/prompts.py`. The advisory policy serves as a generic-assistant baseline, permitting imperatives and targeting three to eight sentences of concise bullet advice. The reflective policy imposes three hard constraints: each response must cite at least one retrieved memory verbatim, avoid imperatives entirely, and end with an open question, within a three- to six-sentence target. Advisory is the baseline against which we measure the Part 1 MIRROR style; a third listening policy is implemented but unused in v0.1.

2.6 Conditions

We compare three memory conditions. The *no_node* baseline presents only the today-question. The *retrieval* condition augments the question with the top-5 neighbors from the embedding index. The *oracle* condition replaces retrieval with every document whose id appears in the item's `evidence_*_ids` lists, capped at k × 3 (= 15) for prompt length. Algorithm 2 gives the test-time pipeline shared by all three.

```
Algorithm 2 — Retrieval-augmented inference
---------------------------------------------------------
input  : item x, SUT m, policy π ∈ {advisory, reflective},
         condition c ∈ {no_node, retrieval, oracle},
         index I, embedder E, top-k = 5
output : response string r

if c = no_node:
    ctx ← ∅
```

```
elif c = retrieval:
    q   ← E.encode(x.question)
    ctx ← top_k(I, q, k = 5)          // cosine similarity
elif c = oracle:
    ctx ← { u ∈ I.docs | u.id ∈ x.evidence_ids }
    ctx ← ctx[0 : 3·k]                 // cap at 15

prompt ← format(π, ctx, x.question)    // policy prompt template
r      ← m.generate(prompt, max_tokens = 200, temperature = 0.3)
return r
```

2.7 Judging

Reflective-quality pairs (advisory vs reflective response) were scored by two cost-tier judges, `openai/gpt-4o-mini` and `anthropic/claude-haiku-4.5`, on four rubrics (specificity, non_directive, emotional_attunement, open_question) with A/B swap for position-bias mitigation. 6 REFL items × 2 SUTs × 2 conditions × 2 judges × 2 swaps = **96 pairwise decisions per rubric** aggregated in `experiments/.../judge_aggregate.json`. Algorithm 3 details the pairwise protocol.

```
Algorithm 3 — Pairwise reflective-quality judging
---------------------------------------------------------
input  : REFL items X_REFL, SUT pool S, judges J,
         rubrics R = {specificity, non_directive,
                      emotional_attunement, open_question}
output : win counts W[s, c, rubric] → {adv, refl, tie}

for each x ∈ X_REFL, s ∈ S, c ∈ {retrieval, oracle}:
    r_adv  ← results[x, s, c, advisory].response
    r_refl ← results[x, s, c, reflective].response
    for each j ∈ J:
        for swap ∈ {False, True}:
            (A, B) ← (r_refl, r_adv) if swap else (r_adv, r_refl)
            v      ← j.score(x.question, A, B, R)   // per-rubric
            if swap:
                v ← {A↔B, tie↔tie}                  // invert
            for each ρ ∈ R:
                W[s, c, ρ][v[ρ]] += 1
return W
```

Each rubric therefore accumulates `|X_REFL| · |S| · 2 · |J| · 2` = 6 · 2 · 2 · 2 · 2 = 96 decisions.

2.8 Budget and reproducibility

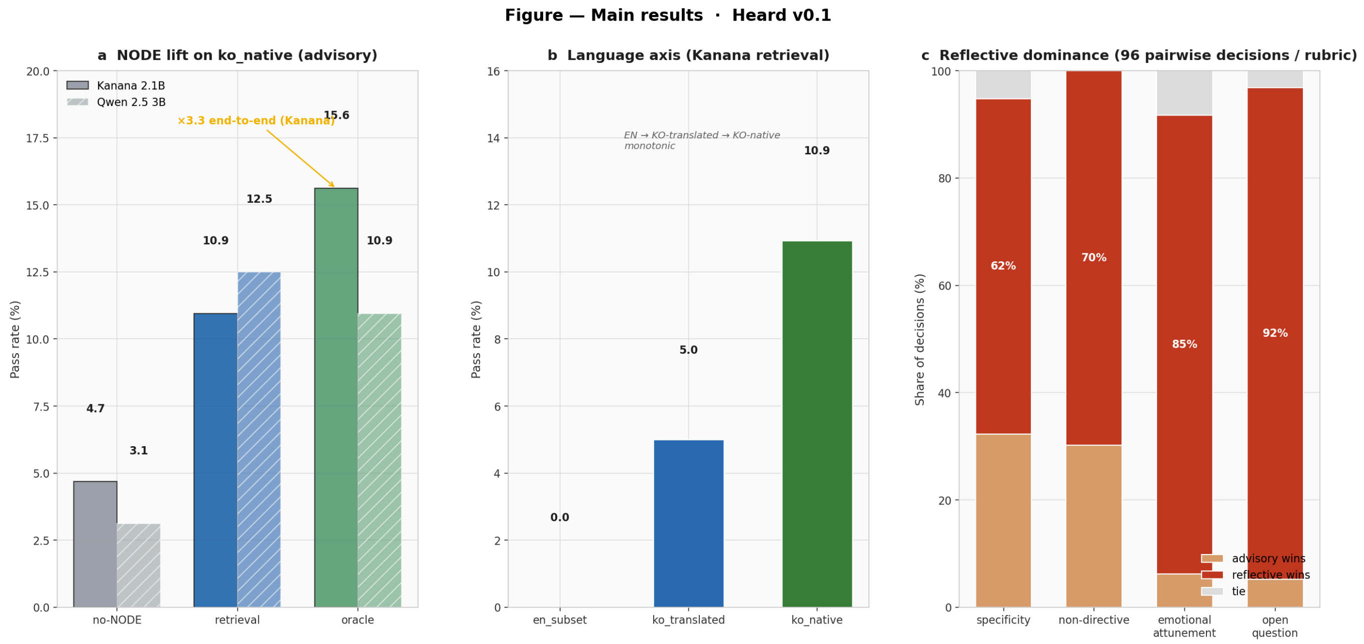
Phase	API cost (USD)
Day 1 utterance corpus	1.88
Day 2 scenarios + translation + 4-gate	2.04
Day 3 judge	0.06

Phase	API cost (USD)
Misc / smoke / retries	~0.02
Total	**≈ 4.00 / 30.00 cap (13%)**

Every API call logs `(model, temperature, seed, usage.cost)` to `experiments/_api_log/api_calls.jsonl`. Diskcache keyed on `(model, messages, temperature, seed, response_format)` makes a cold rerun free. Run artifacts (config, results.jsonl, metrics, per-SUT model revisions) are committed under `experiments/<run_id>/`.

3 Results

The headline results are summarized in the bottom band of Figure 1 and in Figure 2 below; detailed ability and latency analyses appear in Figure 3. Each panel of Figures 2–3 is referenced from the relevant subsection.



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3.1 NODE contribution (ko_native pass rate)

SUT	no_node	retrieval	oracle
kanana_nano (2.1 B)	4.7%	10.9%	15.6%
qwen25_3b (3.0 B)	3.1%	12.5%	10.9%

Kanana shows the monotonic progression we predicted: memory doubles pass rate over a memoryless baseline, and the oracle upper bound extends another 1.4× beyond the retriever. The five-point gap between retrieval and oracle is the retriever's headroom against perfect recall. Denominator is 64 non-REFL items, so Kanana solves 3, 7, and 10 of them as memory conditions strengthen.

Qwen-3B does not follow the same progression; its oracle condition underperforms its retrieval condition (10.9% vs 12.5%). One hypothesis is a summarization bias under long oracle contexts, where Qwen appears to recapitulate the evidence sessions rather than answering the today-question. We return to this in §4.2, where a simple prompt guardrail is the expected fix.

3.2 Language axis (Kanana retrieval)

Track	Pass rate
en_subset	0.0%
ko_translated	5.0%
ko_native	10.9%

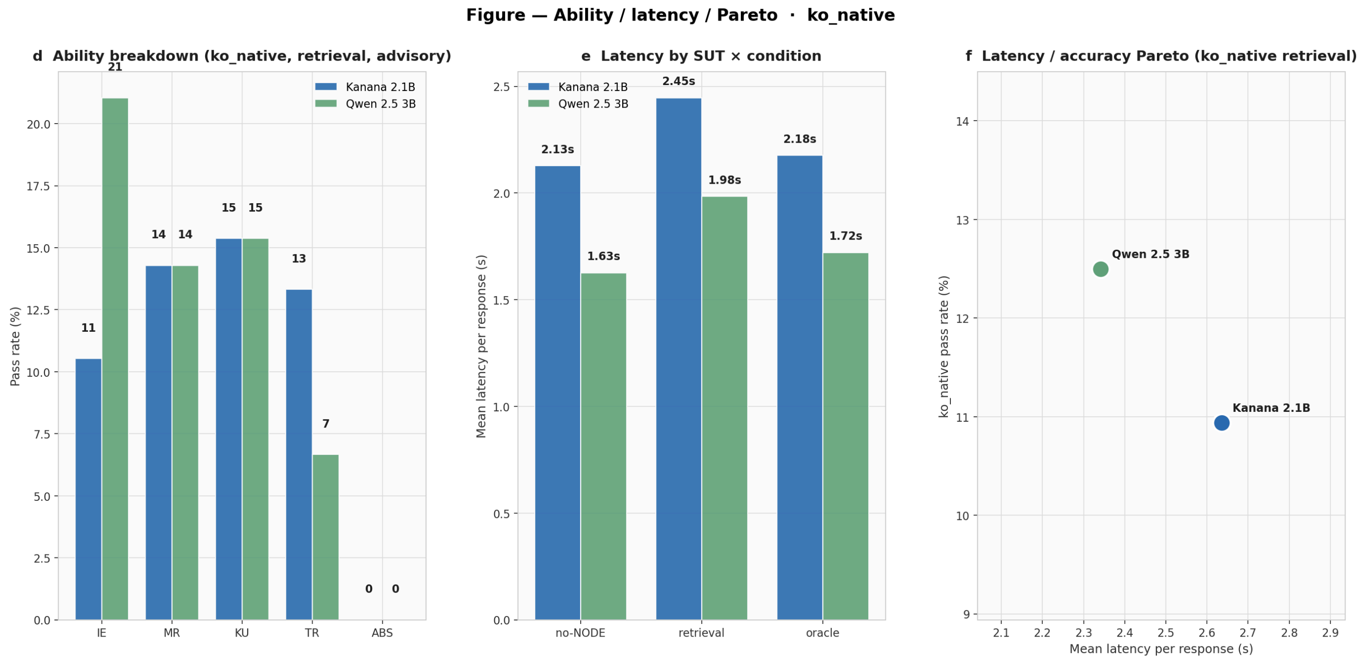
Kanana is a Korean-native 2.1 B SUT, so performance on raw English haystacks (en_subset) collapses even with retrieval; KO-question over EN-haystack (ko_translated) is a midpoint; fully Korean (ko_native) gives the best signal. The progression supports Korean-native data as a requirement rather than a preference.

3.3 Reflective vs advisory (pairwise judge)

Rubric	Advisory wins	Reflective wins	Tie
specificity	31	60	5
non_directive	29	67	0
emotional_attunement	6	82	8
open_question	5	88	3

Summed across both SUTs and both retrieval and oracle conditions, the reflective policy wins every rubric. The advisory-vs-reflective gap is largest on the two rubrics that operationalize the "mirror, do not advise" thesis: emotional attunement at 85% reflective and open-question framing at 92%. Specificity and non-directive win by tighter but still clearly one-sided margins.

3.4 Ability breakdown and latency



{#fig:details}

Per-ability pass rates on `ko_native` retrieval show gains concentrating on factual abilities (IE, TR, KU, and MR), matching our prior that cosine retrieval surfaces named entities well. Abstention (ABS) is uniformly hard: without explicit "no-evidence" signals, the SUT hallucinates rather than refusing, a limitation we return to in §4.3.

Response latency stays under 2.7 s for both SUTs across all conditions on a single L40S, within the typical on-device expectation for a decision-moment assistant. On the latency-accuracy Pareto plane, Qwen-3B sits to the upper-left of Kanana-2.1B for the ko_native retrieval point (higher accuracy, lower latency), suggesting that the larger multilingual model remains competitive on-device when compute is not the bottleneck.

4 Discussion

4.1 Absolute pass rates are low by construction

Adversarial filtering retained only items where three different no-memory baselines failed across three trials. The surviving items are, by construction, hard. Kanana's 10.9% retrieval pass rate on this subset is meaningful evidence of memory contribution; the same SUT on unfiltered questions would look much better on paper and teach less. The headline interpretation of §3.1 is the relative lift (×3.3 end-to-end), not the absolute rate.

4.2 Qwen's oracle anomaly

Kanana improves monotonically from no_node to retrieval to oracle; Qwen-3B's oracle is worse than its retrieval. Qwen's responses under long oracle contexts show a summarization bias, where the model recapitulates the evidence sessions rather than answering the today-question. A concrete fix for v2 is a no-summarization guardrail in the system prompt ("Answer the question. Do not re-narrate the past."). This reads as a prompt artifact rather than a capability ceiling.

4.3 What dense retrieval does not measure

The v0.1 5-ability benchmark measures substring recall, namely whether the SUT reproduces the correct string given retrieved evidence. Dense embedding retrieval handles this well: when the evidence session is in the top-5, substring recall usually succeeds.

Three axes fall outside what this benchmark scores cleanly, each connecting to a Part 1 NODE claim. First, absence queries of the form "the user has never mentioned a vegan menu" resist dense retrieval because cosine similarity always returns some neighbor, encouraging hallucinated confirmation rather than refusal. Second, *entity-aggregation* queries such as "every memory about Grandma Park" require exhaustive graph traversal for completeness, whereas cosine top-k is both approximate and bounded in coverage. Third, *temporal-latest* queries such as "the current permanent-wave product" need exact timestamp filtering, whereas semantic similarity conflates old and new mentions. A NODE-native benchmark of 30 to 50 items targeting these three axes is the natural v2 extension.

4.4 Track B scope limitation

ko_translated keeps the haystack in English (ADR 0003). The "language only" axis is thus partially confounded, because Kanana's ko_translated number reflects both translation and haystack-language effects. Re-running with Korean haystacks is a tractable extension; the cost analysis in ADR 0003 identifies chunked session translation as the right path.

4.5 Validation shortfall

The final ko_native set is 70 items, below the planned 100. Gate 4 (question clarity ≥ 3/5) was the dominant rejection cause: LLM raters are systematically conservative on isolated questions shown without persona context. Giving the rater that context in a future run is a one-line prompt change likely to lift acceptance closer to what a human reviewer would accept.

5 Conclusion

Heard v0.1 contributes, to our knowledge, the first Korean long-term memory benchmark targeting solo-business monologue, built because no existing benchmark meets the three criteria of Korean / monologue / solo-business. On this benchmark, a lightweight on-device retrieval pipeline produces measurable gains for a 2B-parameter Korean SUT (Kanana: 4.7% → 10.9% → 15.6%). A reflective response policy dominates an advisory one on emotional attunement (82/96) and open-question framing (88/96), supporting the Part 1 MIRROR thesis.

Several extensions follow from the v0.1 limitations identified above, ordered by rough value-to-effort priority.

1. A NODE-native capability benchmark of 30 to 50 items targeting absence-strict, entity-aggregation, and temporal-latest queries, the three axes §4.3 argues that dense retrieval cannot reach.
2. Full-haystack Korean translation of ko_translated via chunked session translation (ADR 0003), closing the confound in §4.4.
3. SUT expansion to Kanana-1.5-8B as a reference ceiling and to EXAONE-3.5 and HyperCLOVA-X-SEED for intra-Korean comparison.
4. An int4 quantization axis for tighter on-device simulation.
5. A BM25 baseline to isolate the dense-retrieval advantage.

6 Artifacts and References

6.1 Artifacts

The author website, code repository, and dataset repository links appear in the header of this report. The remaining artifacts used for the results in §3 are located as follows.

	Location
Day 3 main sweep	`experiments/20260423_1610_day3_sweep/`
Per-call API log	`experiments/_api_log/api_calls.jsonl`
ADRs	`docs/decisions/0001-0004*.md`

6.2 References

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7 Acknowledgments and AI Use

Claude Code was used as a coding assistant for implementing the pipeline and scripts in this repository. Figure 1 (`fig_overview.jpg`) was drawn in Paperbanana. All design decisions, including research framing, dataset methodology, hypothesis formulation, result interpretation, and writing direction, were performed by the author.